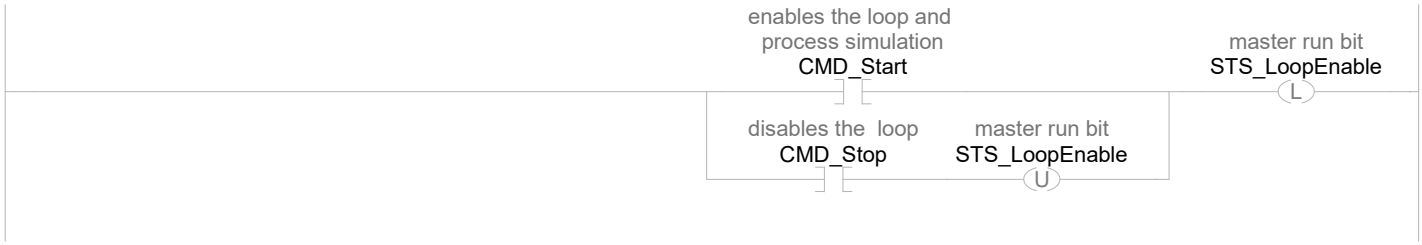
0


enables the loop and
process simulation
CMD_Start
master run bit
STS_LoopEnable
L
disables the loop
CMD_Stop
master run bit
STS_LoopEnable
U

1

Initialize all process and controller support values once at startup so the project always begins from a repeatable state

one-time initialization complete
STS_FirstScanDone

simulated temperature process variable
MOVE
Source 70.0
Dest TIC_PV
70.0

the operator-entered desired setpoint
MOVE
Source 150.0
Dest TIC_SP_Target
150.0

the smoothed setpoint you actually feed into the PIDE
MOVE
Source 70.0
Dest TIC_SP_Ramped
70.0

manual output command value when in manual mode
MOVE
Source 0.0
Dest TIC_ManualCV
0.0

room temp baseline
MOVE
Source 70.0
Dest SIM_AmbientTemp
70.0

how strongly CV affects the heater
MOVE
Source 3.0
Dest SIM_HeaterGain
3.0

lag/time constant of the simulated process
MOVE
Source 45.0
Dest SIM_TauSeconds
45.0

task period in seconds
MOVE
Source 0.1
Dest SIM_TaskDT
0.1

degrees per second allowed
MOVE
Source 1.0

Dest CALC_SP_RampRatePersec
1.0

one-time initialization complete
STS_FirstScanDone
L

loop is enabled

2

master run bit
STS_LoopEnable

loop is ready to run
STS_LoopReady

calc_sp_rampstep = calc_sp_ramprate persec * sim_taskDt

3

loop is ready to run
STS_LoopReady

movement allowed per scan

MUL
Source A CALC_SP_RampRatePersec
1.0
Source B SIM_TaskDT
0.1
Dest CALC_SP_RampStep
0.1

calc_sp_delta = tic_ sp_target - tic_sp_delta

4

loop is ready to run
STS_LoopReady

Target - Ramped

SUB
Source A TIC_SP_Target
150.0
Source B TIC_SP_Ramped
70.0
Dest CALC_SP_Delta
0.0

calc_sp_absdelta = ABS(CALC_SP_DELTA

5

loop is ready to run
STS_LoopReady

absolute difference used for comparison logic

ABS
Source CALC_SP_Delta
0.0
Dest CALC_SP_AbsDelta
0.0

CALC_SP_RampStep_Neg = -CALC_SP_RampStep

6

loop is ready to run
STS_LoopReady

NEG
Source CALC_SP_RampStep
0.1
Dest CALC_SP_RampStep_Neg
0

7

if delta is bigger than one ramp step, incerase TIC_SP_Ramped

loop is ready to run
STS_LoopReady

Target - Ramped

GT	
Source A	CALC_SP_Delta 0.0
Source B	CALC_SP_RampStep 0.1

the smoothed setpoint you actually feed into the PIDE

ADD	
Source A	TIC_SP_Ramped 70.0
Source B	CALC_SP_RampStep 0.1
Dest	TIC_SP_Ramped 70.0

8

if delta is more negative than one ramp step, decrease TIC_SP_Ramped

loop is ready to run
STS_LoopReady

Target - Ramped

LT	
Source A	CALC_SP_Delta 0.0
Source B	CALC_SP_RampStep 0.1

the smoothed setpoint you actually feed into the PIDE

SUB	
Source A	TIC_SP_Ramped 70.0
Source B	CALC_SP_RampStep 0.1
Dest	TIC_SP_Ramped 70.0

9

if the remaining difference is smaller than one step, force the ramped sp to exactly equal the target sp

loop is ready to run
STS_LoopReady

absolute difference used for comparison logic

LE	
Source A	CALC_SP_AbsDelta 0.0
Source B	CALC_SP_RampStep 0.1

the smoothed setpoint you actually feed into the PIDE

MOVE	
Source	TIC_SP_Target 150.0
Dest	TIC_SP_Ramped 70.0

10

forces the pide instruction into program control so mode transitions and output behavior are controlled by ladder logic. Thjis prevents operator or external ownership from override Auto/Manual switching during the simulation

TIC_101_PIDE.ProgProgReq

11

clears the operator control request bit to ensure Program control ownership remains active. this guarantees the loop responds only to Program Auto and program manual commands issued by the project logic.

TIC_101_PIDE.ProgOperReq
U

12

generates one-scan Program Auto request when the auto command transitions from off to on. This switches the pide instruction into automatic mode where controller output is calculated from setpoint and process variable

requests Auto mode
CMD_Auto

ONS_AutoReq
ONS

TIC_101_PIDE.ProgAutoReq
L

13

Clears the Program Auto request bit after the auto pushbutton is released. This ensures the request behaves as a pulse instead of a continously asserted command

requests Auto mode
CMD_Auto
/

TIC_101_PIDE.ProgAutoReq
U

Generates a one-scan Program manual request when the manual command transitions from off to on. This places the PIDE instruction into manual mode allowing direct control of the output using the manual CV value

14 requests manual mode
CMD_Manual — TIC_101_PIDE.ProgManualReq (L)

Clears the program manual request bit after the manual pushbutton is released. This prevents the request from remaining latched and ensures proper mode transition behavior

15 requests manual mode
CMD_Manual (/) — TIC_101_PIDE.ProgManualReq (U)

Writes the ramped setpoint value into the PIDE program setpoint input. The ramped setpoint is used instead of the target setpoint to prevent abrupt changes and improve loop stability during tuning and normal operation

16 loop is ready to run
STS_LoopReady

MOVE
Source TIC_SP_Ramped 70.0
Dest TIC_101_PIDE.SPProg 0.0

Writes the manual controller output value into the PIDE program control variable input. This value is used when the loop is in Program manual mode to allow direct operator control of the output

17 loop is ready to run
STS_LoopReady

MOVE
Source TIC_ManualCV 0.0
Dest TIC_101_PIDE.CVProg 0.0

Copies the calculate PIDE control variable output into a project tag for simulation, trending, and performance analysis of the temperature control loop.

18 loop is ready to run
STS_LoopReady

actual controller output copied from PIDE

MOVE
Source TIC_101_PIDE.CV 0.0
Dest TIC_CV 0.0

19 master run bit
STS_LoopEnable

JSR
Routine Name PlantSim_ST

JSR
Routine Name PIDE_FBD

(End)